AI Gateways for EuroHPC AI Factories and Antennas: Functions, Tools and the SAIFA Example

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***Abstract*—The EuroHPC AI Factories initiative aims to build a network of AI supercomputing centres in Europe, and AI Factory Antennas are to bring their services to countries that do not host an AI Factory. Users will reach these services through several separate access points, and an AI gateway can bring them together. This paper defines ten groups of functions that such a gateway needs, compares them with three existing solutions, Waldur, Open OnDemand and the EuroHPC Federation Platform, and lists possible open-source tools. It then presents an architectural proposal for the *AI gateway* of the Serbian AI Factory Antenna (SAIFA). What is new for an Antenna is that one gateway has to work both with local resources, which it manages, and with the infrastructure of independent partner Factories (Pharos and IT4LIA), which it does not control. The proposal specifies which tasks the gateway performs itself and which system decides on access, allocation and accounting. It is based on the SAIFA design and has not yet been implemented or evaluated.**

***Index Terms*—AI Factory, AI Factory Antenna, AI gateway, EuroHPC, federation, high-performance computing, MLOps, science gateway.**

I. Introduction

Modern artificial intelligence (AI) research needs large computing resources. To give such resources to more users, the EuroHPC Joint Undertaking (JU) started the AI Factories initiative. By October 2025, 19 AI Factories had been selected. They are being set up gradually, and each of them is to host an AI-optimised supercomputer and to offer services such as data management, help with model development, training courses and expert support [1], [2]. In October 2025, 13 AI Factory Antennas were also selected to bring these services to more countries. Each Antenna is linked with one or more AI Factories [3], [4].

Access to large computing resources has long been difficult for many users. Grid and high-performance computing (HPC) systems required command-line tools, certificates and different procedures at each site, and science gateways were developed to give users simple web access to them [5], [6]. The AI Factory network may repeat this problem, because its services will be reached through many separate access points. Startups, small and medium-sized enterprises (SMEs), researchers and public institutions may have to use different portals, accounts and application procedures before they can run one experiment. A gateway for AI services also has to support the whole model development process, safe sharing of data and models, and access to large language models (LLMs).

In this paper, we propose an architecture for the AI gateway of the Serbian AI Factory Antenna (SAIFA) [7], [8], which is linked with Pharos and IT4LIA, and compare it with existing solutions. The function groups, the comparison with existing solutions and the candidate tools are relevant to AI gateways in general, while the architecture is proposed for SAIFA. The architecture has not yet been implemented or tested. Antennas were introduced only in 2025, and we are not aware of an existing gateway designed for their role. The SAIFA gateway manages quotas and accounting for local resources, but it also has to work with the resources of independent partner Factories, which keep their own platforms and access rules. For each function, we state whether the gateway, an existing system or a partner is responsible for it, and which system decides on access, allocation and accounting.

II. Access Points in the AI Factory Network

A. AI Factories

AI Factories will be run by national host organisations around EuroHPC supercomputers. The JU lists the services they are to offer [9], including expert help, AI-optimised computing, data management, training and AI software. Computing time is given through EuroHPC access calls in four modes [10]: *Playground* and *Fast Lane* for startups and SMEs, *Large Scale* for work above 50,000 GPU hours, and *AI for Science and Collaborative EU Projects* for scientific and public sector users. Each Factory will also have its own contact point and its own rules for users. For example, Pharos in Greece plans a Front Desk as the central contact point for users [11].

B. AI Factory Antennas

Antennas are to extend the services of the AI Factories [4]. An Antenna is expected to offer a national entry point, smaller local computing resources, AI test environments, help for specific sectors and training. For large jobs, an Antenna will use the resources of its linked Factories. One Factory can

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work with several Antennas and vice versa. For example, Pharos is linked with four Antennas (Pharos-CY, CALYPSO, VEZILKA and SAIFA), and the Swiss Antenna HEARTS with four Factories [3].

C. Shared European platforms

The first version of the EuroHPC Federation Platform (EFP) was released in April 2026 to reduce differences in access procedures between EuroHPC sites. It gives users one entry point and one login for several EuroHPC systems. It also supports resource requests, data transfer and a shared software catalogue. AI Factories are not yet part of it and will be connected in later versions [12]. The AI-on-Demand Platform is a European catalogue where users can find AI resources [13]. Table I summarises all access points.

TABLE I
ACCESS POINTS IN THE AI FACTORY NETWORK

Access point	What it offers to the user
EuroHPC access calls	Applying for computing time (four modes for industry and science)
EuroHPC Federation Platform	One login, common resource allocation and data transfer, shared software catalogue for EuroHPC systems
AI Factory contact office	Consulting, expert help, training, data services and computing at a national site
AI Factory Antenna	National entry point, local computing, help for new users, sending users and results to linked Factories
AI-on-Demand Platform	European catalogue of AI resources

These access points cover different needs, and the SAIFA gateway is meant to connect to them rather than replace them. It sends users and jobs to the partner Factories, can make approved resources discoverable in the AI-on-Demand Platform [8], and should reuse EFP login and allocation once AI Factories join the platform.

III. Proposed Architecture of the SAIFA AI Gateway

The proposed gateway works as *middleware*. It receives requests from users or applications, checks all of them against common rules for identity, access and metadata, and routes each request to the appropriate service. Fig. 1 shows its main parts. The actual work is still done by existing systems, such as schedulers, storage and model servers, and by partner platforms.

The ten function groups in Table II come from the SAIFA design. Three rules apply to all of them: every action is checked against the user's permissions, every dataset and model has a version and a record of its origin (lineage), and every job is checked against the quota before it is queued.

The split of responsibilities follows the SAIFA design [8]. The institutional or EuroHPC identity provider authenticates the user, and the gateway links this identity to its own account. For local resources, quota and accounting logic belongs to the gateway, which keeps the record of available quota,

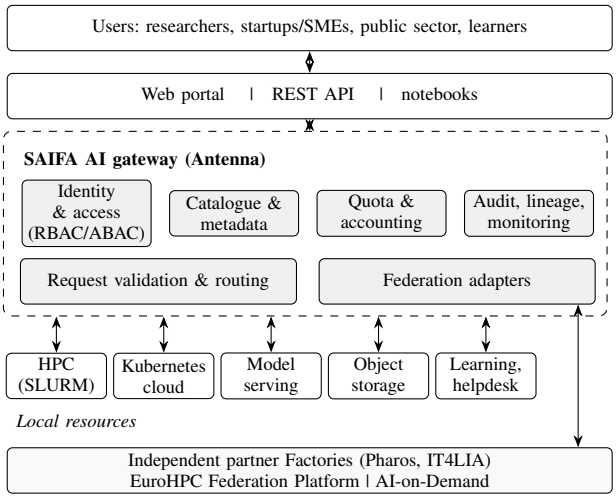


Fig. 1. Proposed architecture of the SAIFA AI gateway between users, local resources, partner Factories and European services.

TABLE II
MAIN FUNCTION GROUPS AND THEIR COVERAGE BY EXISTING SOLUTIONS

Group	Main features	W	O	E
Identity & access	Institutional (eduGAIN) and EuroHPC sign-in, account linking, RBAC/ABAC, scoped API tokens, revocation	◐	◐	●
Organisations & projects	Workspaces, membership, roles, resource visibility (private, project, platform, public)	●	○	◐
Catalogue	Search of models, datasets, workflows, services and courses; source attribution of federated entries	◐	○	◐
Data & model management	Resumable upload, versioning, model and data cards, review and publication workflow	○	◐	◐
Compute jobs	Job templates, quota reservation, routing to HPC or Kubernetes, status, logs, result retrieval	◐	●	●
Inference	Authenticated, rate-limited access to served models and LLMs	○	○	○
Experiments & MLOps	Experiment tracking, run comparison, benchmark evaluation, lineage across job, dataset and model	○	○	○
Training	Courses, learning paths, preconfigured notebook labs with controlled resources	○	◐	○
Operations	Monitoring, usage accounting, audit trail, notifications, contextual helpdesk tickets	◐	○	◐
Federation	Catalogue exchange, metadata mapping, identity brokering, cross-Factory execution and accounting	◐	○	◐

W: Waldur [14], O: Open OnDemand [6], [15], E: EFP [16]. ● most listed features in the standard release (configuration only), ◐ some available, the rest needs extension or integration, ○ not available (new development).

reserves quota before a job is queued and records inference consumption. The gateway stores metadata, review decisions, job records, audit logs and lineage, but job status always comes from the scheduler, locally or at the partner site. Some details, such as the quota hierarchy, review states and the split of tasks with system operators, are not final yet [8].

IV. Description of the Main Functions

A. Login, access and collaboration

Users log in with their university or institute account (for example through eduGAIN) or with a EuroHPC account.

TABLE III
POSSIBLE TOOLS FOR EACH PART AND WORK NEEDED FOR THE GATEWAY, BASED ON [8]

Part	Main option(s)	Work needed for the gateway
User access		
Portal & UX	Next.js/React	Unified UI, backend-for-frontend, tool embedding
Identity & policies	Keycloak; Open Policy Agent	Role mapping, ABAC attributes, account linking, policy model
Notebooks	JupyterHub + KubeSpawner	Launch from portal, quotas, dataset mount, audit
Data and models		
Catalogue & search	CKAN or Dataverse; OpenSearch + pgvector	Unified catalogue, access filtering in index, federation
Model registry & experiments	MLflow	Model cards, approval workflow, access rights
Storage	PostgreSQL, S3 storage on MinIO/Ceph, Valkey	Metadata links, lifecycle, retention
Lineage	OpenLineage + Marquez	Links between jobs, datasets, models and results
Compute and inference		
LLM inference & serving	vLLM; KServe	API proxy, quotas, routing, audit
Fine-tuning	Hugging Face TRL/PEFT	Approved templates, HPC submission, result registration
HPC & batch	SLURM, Open OnDemand; Kubernetes Jobs	HPC bridge, unified job model, SLURM/Kubernetes routing
Workflows & containers	Argo Workflows; Harbor + Trivy, Apptainer on HPC	Workflow templates, image approval and HPC conversion
Operations and federation		
Observability & audit	Prometheus, Grafana, Loki, OpenTelemetry; append-only log + write-once archive	Correlation with user/job, shared audit events
Learning & support	Moodle; Zammad or GLPI	Portal links, lab quotas, contextual tickets
Federation & accounting	DCAT/DCAT-AP adapters; Waldur as cross-layer option	Partner adapters, per-Factory accounting

Access is decided by the user's role (role-based access control, RBAC) and by other attributes, such as organisation, project and type of data (attribute-based access control, ABAC) [17]. Seeing a resource in the catalogue and using it are two separate permissions, and search shows resources of other projects only if they are public or their owners allow it.

B. Datasets and models

To train a model, a user only needs access to the dataset and can repeat runs as often as needed. Each result is linked automatically to its dataset, settings and job. Formal review, in which a data administrator or model reviewer checks the metadata, origin, licence and test results, is needed only when a dataset or model is published, so that it does not slow down everyday work [8]. A dataset or model version is published only after the reviewer approves it, and the decision is recorded. Only published resources can be shared with partners. Large files are uploaded in resumable parts with checksums, and a stored version cannot be changed. Where appropriate, published resources get FAIR metadata [18], which eases sharing with partner catalogues.

C. Jobs, inference and experiments

Before a job is queued, the gateway checks the input data and the user's access, and reserves quota. After that, it sends the job to HPC through SLURM [19] or to Kubernetes [20]. Sensitive data, such as medical records, can stay where it is, and the job is sent to the data. The gateway collects status, logs and results and shows them in the same way for all systems. Containers are converted to Apptainer/Singularity images [21] for HPC. Requests to a model go through a proxy that authenticates the user, checks the limits and records the usage. When runs are compared, a person always decides which model to keep.

D. What is new for an Antenna: local and partner resources

A Factory mostly *publishes* its services and resources to European platforms. For an Antenna the task is harder, because besides its local resources it also uses those of partner Factories, which are independent and decide on the use of their own resources. The gateway only passes the user's identity on to the partner, translates metadata into the partner's format, sends large jobs to the partner site, brings the results back to the user's project and keeps usage separately for each Factory. Each partner is connected through its own adapter, for example using DCAT [22], so partners do not have to adopt the Antenna's platform. The gateway shows where each partner resource comes from and never edits it. At first, the links to partner Factories are read-only or simulated. If a partner is unavailable, the gateway still works for local users.

V. EXISTING SOLUTIONS AND POSSIBLE TOOLS

Three existing solutions already cover part of the proposed architecture. Waldur is a platform for projects, allocation and accounting on cloud and HPC systems [14], and Open OnDemand is a web portal for HPC centres [6]. The EFP [12] is itself built on MyAccessID, Waldur and Open OnDemand, among other components [23]. Table II shows what each of them offers to users. We assigned the marks from the public documentation of each solution (October 2026), without hands-on tests. For the EFP, we rated what the platform offers to its users, not what its components could provide. Together they cover login, projects, allocation, accounting and interactive HPC work. They do not provide the AI-specific functions (dataset and model review, inference with quotas, experiment lineage). Waldur can connect to other Waldur deployments, but none of the three is built for a gateway that combines local resources with the different platforms of partner Factories and keeps usage separately for each of them.

The proposed architecture adds these functions and common rules for access, quotas and metadata.

Existing tools are used first, and new ones are built only when needed. Table III gives the main options for each part and the work that the SAIFA gateway still has to do itself; alternatives are listed in [8]. Most options are well known open-source projects, for example JupyterHub [24], [25], MLflow [26], vLLM [27] and Open OnDemand [6].

Tools are chosen in two steps. Each tool must first meet the requirements for its area, and those that pass are compared by ease of use, integration and operating effort, cost, maturity and legal issues, with a preference for open-source tools on local servers [8].

VI. SAIFA Context and Planned Validation

SAIFA, the Serbian Antenna linked with Pharos and IT4LIA, runs from May 2026 to April 2029 [7]. Its local resources are the PARADOX HPC cluster at the Institute of Physics Belgrade and the AI computing infrastructure of the Office for IT and eGovernment. The function groups in Table II are based on 23 user stories, 18 from its four domains (language and culture, energy and manufacturing, health, environment) and 5 cross-domain. Examples include flood prediction, synthetic genome data and hate-speech detection in Serbian. From these stories, the design derives 20 representative system use cases, which still have to be validated with domain owners [8].

The proposal will be validated in two steps. First, a basic proof-of-concept setup will test the candidate tools from Table III together, including where the data is stored, how identity is passed on, how resource use is charged and what happens when a service fails [8]. Second, the beta version planned for April 2027 must let a user log in, find a resource, run a job and follow its progress.

VII. Conclusion

This paper described the main functions of an AI gateway and presented an architectural proposal for the SAIFA gateway, which has not yet been implemented or evaluated. Compared with Waldur, Open OnDemand and the EFP, most new development will likely be needed for the AI-specific functions and for the work with partner Factories, including separate accounting for each of them. Open questions for future work are how gateways will use the EFP for login and allocation, which common metadata to use for models and datasets, and which data rules apply when jobs run in another country.

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